

Q: Transformer-based Object Detection on Pascal VOC Dataset using DETR

You are tasked with training and evaluating an object detection model using the **DETR (DEtection TRansformer)** architecture on a **public dataset** such as **Pascal VOC** or **COCO (subset)**.

You may use Pascal VOC 2007 or a small COCO dataset (e.g., COCO-2017 val subset) for faster execution.

Tasks:

1. Dataset Handling:

- Download and load the **Pascal VOC 2007** dataset using `torchvision.datasets.VOCDetection`.
- Preprocess the data:
 - Resize the image so that the longest side is 800 pixels.
 - Normalize with mean `[0.5, 0.5, 0.5]` and std `[0.5, 0.5, 0.5]`.
 - Convert annotations to match DETR format (list of dicts with boxes and labels).

2. Model Setup:

- Load a pre-trained **DETR model** (`facebook/detr-resnet-50`).
- Modify the classification head to support the number of Pascal VOC classes (20).
- Train or fine-tune DETR on Pascal VOC training split for at least **5 epochs**.
- Use appropriate loss functions (Hungarian loss, bbox loss).

3. Evaluation:

- Run inference on the Pascal VOC validation set.
- For each image, draw predicted bounding boxes and class labels.
- Calculate detection metrics:
 - **mAP (mean Average Precision)**
 - **IoU (Intersection over Union)** threshold ≥ 0.5 .
- Compare predictions to ground truth visually and numerically.

4. Attention Visualization (Optional Bonus):

- Extract and visualize the decoder self-attention or cross-attention map for a single detected object.
- Interpret how DETR attends to object locations compared to CNN-based detectors.

5. Final Report:

Submit a report with:

- Description of dataset and preprocessing steps.
- Model architecture summary.

- Training and validation performance.
- Example outputs with bounding boxes.
- Brief explanation: “How does DETR differ from CNN-based object detectors like Faster R-CNN?”